



5-DAY CAPSTONE PROJECT

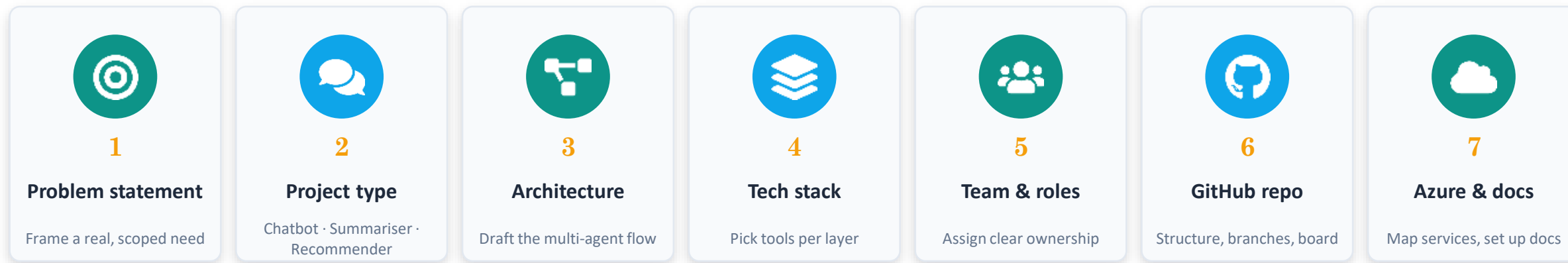
Day 1 — Foundation & Planning

Guidelines for scoping, designing and launching an agentic-AI build

Data Mining • Agentic AI • Multi-Agent Architecture • Azure

Day 1 at a glance

Seven deliverables move you from a blank page to a launch-ready repository.



Goal of Day 1: lock scope before you write code. Every later day depends on decisions made today — a vague problem statement costs you on Day 4, not Day 1.



DAY 1 · STEP 1

Define the problem statement

A sharp problem statement is the single biggest predictor of a successful capstone.

What a strong statement contains

- 1 **User & context** — Who feels the pain, and where?
- 2 **The need / pain** — One concrete problem, not three
- 3 **Proposed solution** — What you'll build, in a sentence
- 4 **Success measure** — How you'll know it worked
- 5 **Data source** — What real or synthetic data feeds it
- 6 **Scope fence** — What is explicitly out of scope for 5 days

Fill-in template

For **[users]** who **[need]**, we build a **[chatbot / summariser / recommender]** that **[value]**, measured by **[metric]**. Unlike [today's workaround], it uses agents to **[reason / retrieve / act]**.

Avoid these traps

- “Boil-the-ocean” scope — pick one workflow
- No data plan — confirm data exists before Day 2
- Unmeasurable goals like “make it smart”
- Solution-first framing that hides the real need



DAY 1 · STEP 2

Choose your project type

Three proven archetypes — each maps cleanly onto a multi-agent pattern.



Chatbot

Conversational Q&A or task assistant

Agentic angle

- RAG over your knowledge base
- Tool-calling agents for actions
- Great for support, onboarding, internal help



Summariser

Condense long or messy content

Agentic angle

- Map-reduce / refine over chunks
- Vector DB for retrieval at scale
- Great for reports, meetings, research



Recommender

Suggest the next best item or action

Agentic angle

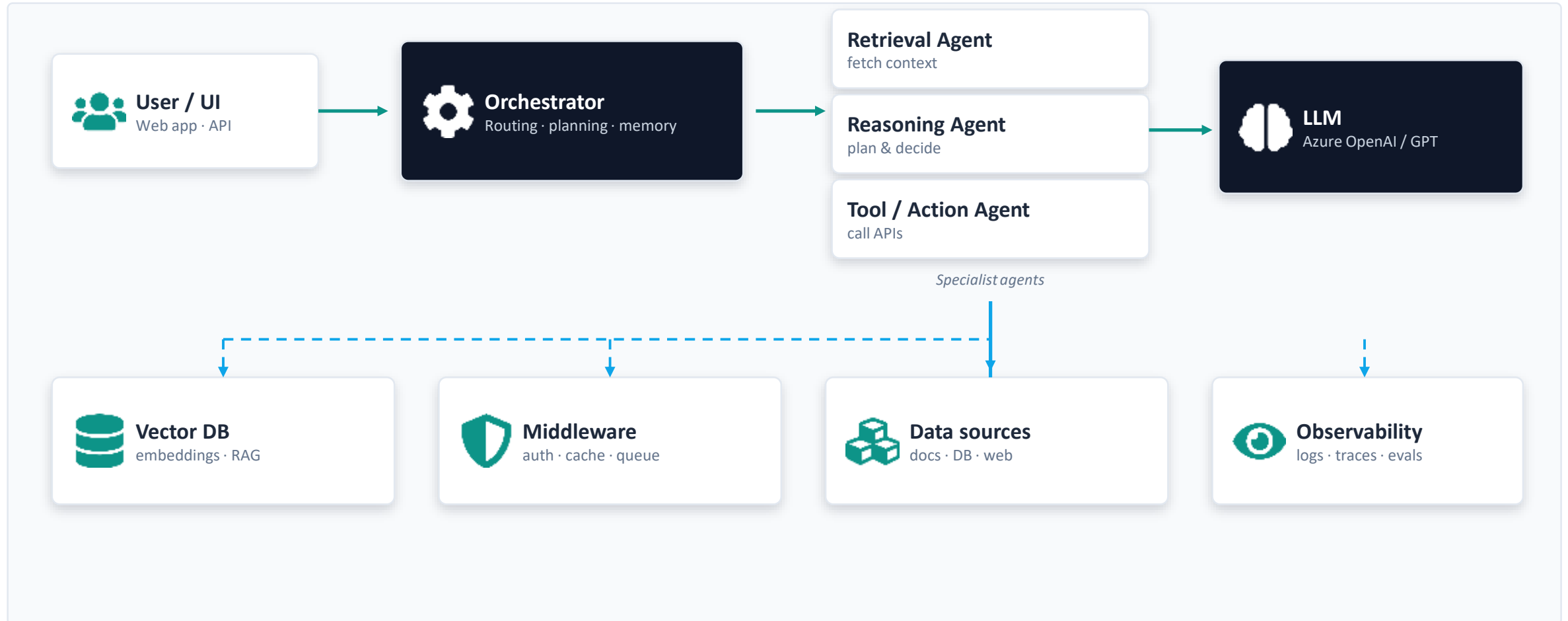
- Embeddings + similarity search
- Agent re-ranks with business rules
- Great for content, products, learning



DAY 1 · STEP 3

Draft the architecture diagram

A reference multi-agent flow — adapt the boxes, keep the layers.





DAY 1 · STEP 4

Select your tech stack

Decide one tool per layer — favour what the team already knows.



Frontend / UI

Streamlit · React · Next.js · Gradio



Orchestration

LangGraph · LangChain · Semantic Kernel · AutoGen · CrewAI



LLM / Models

Azure OpenAI (GPT-4o) · Anthropic Claude · open-weights via vLLM



Vector DB

Azure AI Search · Pinecone · pgvector · Chroma · Qdrant



Middleware

FastAPI · Redis cache · Service Bus / queues · API gateway



Observability

LangSmith · Azure App Insights · OpenTelemetry · Langfuse



Deployment

Azure Container Apps · Functions · Docker · GitHub Actions



DAY 1 · STEP 5

Team formation & role assignment

Every role has a clear owner — small teams double up, nobody is idle.



Team Lead / PM

Owns scope, timeline & daily standup; unblocks the team



Solution Architect

Owns the architecture diagram & key design decisions



Agent / Backend Dev

Builds orchestrator, agents and tool integrations



Data / Vector Engineer

Owns data pipeline, embeddings and the vector DB



Frontend Dev

Builds the UI and the demo experience



DevOps / Observability

Owns repo, CI/CD, Azure deploy, logging & evals

Tip: agree a daily 15-min standup and write roles into the README so ownership is visible.



DAY 1 · STEP 6

Set up the GitHub repository

One source of truth from hour one — structure, branches and a board.

Recommended structure

```
capstone/
├── README.md
├── docs/           architecture, decisions
├── src/
│   ├── agents/     agent definitions
│   ├── orchestrator/
│   ├── retrieval/  vector + RAG
│   └── api/        FastAPI / endpoints
├── ui/            frontend
├── infra/         Azure / IaC, Dockerfile
├── tests/
├── .github/workflows/ CI
├── .gitignore     .env.example
└── requirements.txt
```



Branching

main protected; feature branches + PR review before merge



Issues & board

Turn the 5-day plan into issues on a GitHub Project board



Secrets

Never commit keys — use .env.example + GitHub Secrets



CI from day 1

Actions: lint + tests on every PR so main always runs



DAY 1 · STEP 7

Azure services mapping

Map each architecture component to its Azure equivalent.

Component	Azure service	What it gives you
LLM / models	Azure OpenAI Service	GPT-4o, embeddings, content safety
Vector DB / RAG	Azure AI Search	Vector + hybrid search, semantic ranker
Orchestration host	Azure Container Apps	Run agents & API; scales to zero
Serverless logic	Azure Functions	Event-driven tools & glue code
State / memory	Azure Cosmos DB	Chat history, sessions, metadata
Middleware / queue	Azure Service Bus	Decouple agents, async tasks
Secrets	Azure Key Vault	Store API keys & connection strings
Observability	Azure Monitor + App Insights	Traces, metrics, dashboards, alerts



DAY 1 · STEP 7

Documentation structure

Docs written today save hours of confusion on Day 5.



README.md

Project pitch, setup steps, how to run, team & roles



docs/architecture.md

Diagram + component walkthrough & data flow



docs/decisions.md

ADRs: each key choice, options weighed, why



docs/data.md

Data sources, schema, embedding & chunking strategy



docs/api.md

Endpoints, request/response examples



docs/demo.md

Demo script + evaluation results for Day 5

Living docs: update them as decisions change — stale docs are worse than none.

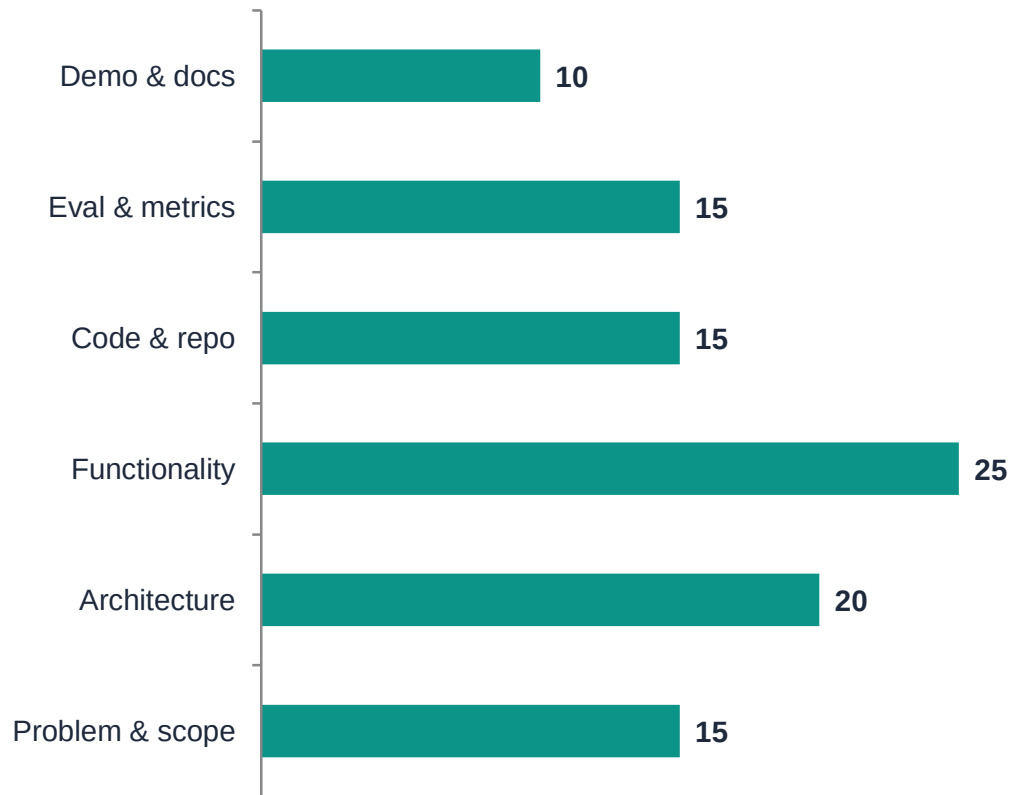


DAY 1 · STEP 8

Evaluation metrics — scoring rubric

How the 5-day capstone is judged · 100 points total.

Weighting



What each band rewards

15

Problem & scope Clear, real, well-scoped problem & success metric

20

Architecture Sound multi-agent design; right tools per layer

25

Functionality Works end-to-end on the core use case

15

Code & repo Clean, organised, reproducible; good git hygiene

15

Eval & metrics Defined metrics + evidence the system meets them

10

Demo & docs Convincing demo and complete documentation

The 5-day milestone track

Daily checkpoints — used alongside the rubric to keep teams on pace.



Lock scope today. Build with confidence all week.